

An extreme-boundary characterization of non-SPR isometric embeddings into $C(K)$

Candidate full solution to Question 6.6 of arXiv:2210.05114

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Status: candidate full solution; human review required. For real Banach spaces, this packet gives an intrinsic necessary-and-sufficient criterion for the existence of an isometric embedding into some $C(K)$ whose range fails stable phase retrieval. It also identifies the sharp worst-case SPR constant over all isometric $C(K)$ -representations.

1 Source and exact question

The source is D. Freeman, T. Oikhberg, B. Pineau, and M. A. Taylor, *Stable phase retrieval in function spaces*, arXiv:2210.05114v1, submitted 11 October 2022 and subsequently published in *Mathematische Annalen* 390 (2024) [1]. Unless explicitly stated otherwise, the source works over the real field. Question 6.6 is on page 48 of the arXiv PDF:

In light of Proposition 6.5, we ask:

Question 6.6. Which Banach spaces E isometrically embed into $C(K)$ in a non-SPR way?

Figure 1: Question 6.6, rendered from page 48 of the source PDF.

Transcription. “Which Banach spaces E isometrically embed into $C(K)$ in a non-SPR way?”

Immediately after the question, the source proves a three-dimensional example for which every isometric $C(K)$ -embedding is SPR. Its Lemma 6.8 supplies the one-sided extreme-point estimate that motivates the answer below:

2 Definitions

Throughout, E is a nonzero real Banach space. If $J : E \rightarrow C(K)$ is a linear isometric embedding, define its *overlap constant*

$$\text{ov}(J) := \inf_{x,y \in S_E} \| |Jx| \wedge |Jy| \|_\infty.$$

The range $J(E)$ does C -stable phase retrieval (C -SPR) when

$$\min\{\|Jx - Jy\|_\infty, \|Jx + Jy\|_\infty\} \leq C \| |Jx| - |Jy| \|_\infty \quad (x, y \in E).$$

The following lemma is needed for the proof, and may be of interest in its own right.

Lemma 6.8. *Suppose K is a Hausdorff compact, E is a Banach space, and $J : E \rightarrow C(K)$ is an isometric embedding. Denote by $\mathcal{F}$ the set of all extreme points of the unit ball of E^* . Then, for any $x, y \in E$,*

$$||Jx| \wedge |Jy|| \geq \sup_{e^* \in \mathcal{F}} |e^*(x)| \wedge |e^*(y)|.$$

Figure 2: The statement of source Lemma 6.8, rendered from page 48.

Write $\text{spr}(J)$ for the optimal such constant, with $\text{spr}(J) = \infty$ if the range is not SPR.

Let $\mathcal{E}_E = \text{ext } B_{E^*}$ and introduce the intrinsic *extreme-boundary overlap*

$$\Delta(E) := \inf_{x, y \in S_E} \sup_{e^* \in \mathcal{E}_E} (|e^*(x)| \wedge |e^*(y)|). \quad (1)$$

This number depends only on the isometric Banach-space structure of E ; it does not refer to an ambient function space.

3 Proof Intuition

Every isometric representation $J : E \rightarrow C(K)$ must display every extreme functional of B_{E^*} as a signed point evaluation. Therefore the overlap of Jx and Jy can never be smaller than the simultaneous values forced by those extreme functionals. This is the source's Lemma 6.8.

The missing converse is obtained by using the smallest canonical compactum that contains exactly this information:

$$K_E = \overline{\text{ext } B_{E^*}}^{w*}.$$

Krein–Milman makes the evaluation map $x \mapsto (e^* \mapsto e^*(x))$ an isometry into $C(K_E)$, while weak-star continuity shows that its overlap is *exactly* the extreme-boundary quantity in (1). Finally, a real lattice identity converts overlap into the optimal SPR constant with no loss. Hence $\Delta(E) = 0$ is precisely the requested criterion for a non-SPR isometric embedding.

4 Full characterization

Theorem 1 (Sharp extreme-boundary criterion). *Let E be a nonzero real Banach space and define $\Delta(E)$ by (1). Then:*

- (i) *for every compact Hausdorff K and every linear isometry $J : E \rightarrow C(K)$,*

$$\text{ov}(J) \geq \Delta(E), \quad \text{spr}(J) \leq \frac{1}{\Delta(E)},$$

with the convention $1/0 = \infty$;

(ii) if

$$K_E = \overline{\text{ext } B_{E^*}}^{w^*} \quad \text{and} \quad (J_E x)(e^*) = e^*(x),$$

then K_E is compact Hausdorff, $J_E : E \rightarrow C(K_E)$ is a linear isometry, and

$$\text{ov}(J_E) = \Delta(E), \quad \text{spr}(J_E) = \frac{1}{\Delta(E)}.$$

(iii) consequently, the following are equivalent:

- (a) E admits a non-SPR linear isometric embedding into $C(K)$ for some compact Hausdorff K ;
- (b) the canonical embedding $J_E : E \rightarrow C(K_E)$ is non-SPR;
- (c) $\Delta(E) = 0$.

Moreover,

$$\sup\{\text{spr}(J) : J : E \rightarrow C(K) \text{ a linear isometry, } K \text{ compact}\} = \frac{1}{\Delta(E)}.$$

Thus $\Delta(E) = 0$ is an intrinsic and sharp answer to Question 6.6.

Proof. We separate the proof into three exact ingredients.

1. Overlap and the optimal real SPR constant. For elements u, v of a real vector lattice, the scalar identity applied by lattice functional calculus is

$$||u + v| - |u - v|| = 2(|u| \wedge |v|). \quad (2)$$

Let F be any real normed subspace of a Banach lattice, and put $d_F = \inf_{u, v \in S_F} ||u| \wedge |v||$. If u, v are nonzero and $a = ||u|| \leq b = ||v||$, monotonicity of the lattice infimum and lattice norm gives

$$\begin{aligned} ||u| \wedge |v|| &= a \left\| \frac{|u|}{a} \wedge \frac{|v|}{b} \right\| \\ &\geq a \left\| \frac{|u|}{a} \right\| \geq ad_F. \end{aligned}$$

Hence

$$\min\{||u||, ||v||\} \leq d_F^{-1} ||u| \wedge |v||. \quad (3)$$

Conversely, substitute $f = u + v$ and $g = u - v$ in the SPR inequality. Its left side is $2 \min\{||u||, ||v||\}$, while by (2) its modulus-difference term is $2||u| \wedge |v||$. Thus no SPR constant smaller than $1/d_F$ is possible. Applying (3) to $u = (f + g)/2$, $v = (f - g)/2$ proves the converse inequality. Therefore

$$\text{the optimal real SPR constant of } F \text{ is exactly } 1/d_F, \quad (4)$$

again with value ∞ when $d_F = 0$. In particular, $\text{spr}(J) = 1/\text{ov}(J)$.

2. Every isometric embedding dominates the extreme boundary. We recall and include the short proof of source Lemma 6.8 [1, Lemma 6.8]. Let $J : E \rightarrow C(K)$ be a linear isometry and fix $e^* \in \text{ext } B_{E^*}$. Hahn–Banach gives a measure $\mu \in M(K) = C(K)^*$ with

$$||\mu|| = 1, \quad J^* \mu = e^*.$$

The fibre

$$\mathcal{F}_{e^*} = \{\nu \in B_{M(K)} : J^*\nu = e^*\}$$

is nonempty, weak-star compact, and convex, so it has an extreme point ν . If $\nu = (\nu_1 + \nu_2)/2$ with $\nu_i \in B_{M(K)}$, then

$$e^* = \frac{J^*\nu_1 + J^*\nu_2}{2}.$$

Extremality of e^* in B_{E^*} forces $J^*\nu_1 = J^*\nu_2 = e^*$; extremality of ν in the fibre then forces $\nu_1 = \nu_2 = \nu$. Thus ν is extreme in $B_{M(K)}$. The extreme points of the real measure ball are $\pm\delta_t$ for $t \in K$, so $e^* = \pm J^*\delta_t$ for some t . Consequently, for all $x, y \in E$,

$$\| |Jx| \wedge |Jy| \|_\infty \geq |e^*(x)| \wedge |e^*(y)|.$$

Taking the supremum over e^* and then the infimum over $x, y \in S_E$ gives $\text{ov}(J) \geq \Delta(E)$. Formula (4) proves part (i).

3. The canonical extreme-boundary embedding attains the bound. By Banach–Alaoglu and Krein–Milman [2, Chapters 3–4], $K_E = \overline{\text{ext } B_{E^*}}^{w^*}$ is a nonempty compact Hausdorff space and the extreme points norm E . Therefore

$$\|J_E x\|_{C(K_E)} = \sup_{e^* \in K_E} |e^*(x)| = \sup_{e^* \in \text{ext } B_{E^*}} |e^*(x)| = \|x\|,$$

so J_E is a linear isometry. For fixed x, y , the function

$$e^* \mapsto |e^*(x)| \wedge |e^*(y)|$$

is weak-star continuous. Its supremum over K_E therefore equals its supremum over the dense subset $\text{ext } B_{E^*}$. It follows that

$$\begin{aligned} \text{ov}(J_E) &= \inf_{x, y \in S_E} \sup_{e^* \in K_E} (|e^*(x)| \wedge |e^*(y)|) \\ &= \inf_{x, y \in S_E} \sup_{e^* \in \text{ext } B_{E^*}} (|e^*(x)| \wedge |e^*(y)|) = \Delta(E). \end{aligned}$$

Formula (4) now gives $\text{spr}(J_E) = 1/\Delta(E)$. The equivalences and the sharp supremum statement follow immediately. $\square$

5 Concrete consequences

Corollary 2 (Finite-dimensional form). *If E is finite dimensional, then E admits a non-SPR isometric embedding into some $C(K)$ if and only if there exist $x, y \in S_E$ such that*

$$\text{for every } e^* \in \text{ext } B_{E^*}, \quad e^*(x) = 0 \quad \text{or} \quad e^*(y) = 0.$$

For a polyhedral space this is a finite vertex test on B_{E^} .*

Proof. In finite dimensions, $S_E \times S_E$ is compact. The function

$$(x, y) \mapsto \sup_{e^* \in K_E} (|e^*(x)| \wedge |e^*(y)|)$$

is continuous because it is the maximum of a jointly continuous function over the compact set K_E . Hence the infimum defining $\Delta(E)$ is attained. It is zero exactly under the displayed annihilation alternative, and Theorem 1 applies. $\square$

For example, $E = \ell_\infty^2$ has $\Delta(E) = 0$: take $x = (1, 0)$ and $y = (0, 1)$, while the extreme points of $B_{E^*} = B_{\ell_1^2}$ are the signed coordinate functionals. Thus the identity representation of ℓ_∞^2 is non-SPR. By contrast, the three-dimensional space in source Proposition 6.7 satisfies $\Delta(E) \geq 1/3$ by the calculation in that proposition, so every isometric $C(K)$ -embedding is at worst 3-SPR. These examples illustrate that the criterion detects precisely the phenomenon that uniform non-squareness alone does not classify.

6 Verification notes

The accompanying `verification.md` gives a separate adversarial audit. The main checks are:

1. the result is explicitly real-scalar, matching the source convention;
2. K_E is weak-star compact even when E is nonseparable, and the evaluation functions are continuous;
3. Krein–Milman makes the extreme boundary norming, so the canonical map is genuinely isometric;
4. taking weak-star closure does not change the relevant supremum because the simultaneous-overlap function is continuous;
5. the real lattice identity gives the *optimal* reciprocal constant, not merely one implication;
6. the source’s one-sided lemma applies to every isometric embedding, while the canonical construction proves attainability and the missing converse.

No computational experiment is used as proof. The packet’s only code renders the two source crops from the copied source PDF.

7 Scope, novelty check, and review recommendation

Scope. The theorem fully answers Question 6.6 in the source’s real-scalar setting by an intrinsic isometric criterion. It does not give a classification by Banach-space isomorphism type, and it does not claim the analogous sharp criterion for complex SPR, whose phase group is larger and whose overlap condition is not sufficient.

Bounded novelty check. On 19 July 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2210.05114, Question 6.6, non-SPR isometric embeddings, extreme points, and canonical boundary embeddings; no duplicate was found. Exact-phrase web searches and author/title searches were also performed. The source paper and the later arXiv papers 2504.06693, 2512.08110, and 2512.08114 were checked in full text where relevant. Those later papers treat complex SPR or the distinct Cantor–Bendixson hierarchy Question 6.4; no statement matching Theorem 1 was located. The source already contains the one-sided Lemma 6.8, but it does not state the canonical attainability, necessary-and-sufficient criterion, or optimal worst-case constant proved here. This was a bounded search and does not establish originality.

Human-review recommendation. Treat the packet as a likely valid candidate full answer to the exact real Question 6.6. The highest-value review issue is conceptual: decide whether an intrinsic sharp extreme-boundary criterion counts as the intended “which spaces” classification, or whether the question sought only a description in pre-existing geometric classes. Mathematically, reviewers should audit the fibre-extreme-point argument and the claim that the canonical boundary embedding attains the universal lower bound.

References

- [1] D. Freeman, T. Oikhberg, B. Pineau, and M. A. Taylor, *Stable phase retrieval in function spaces*, arXiv:2210.05114v1 (2022); *Math. Ann.* 390 (2024), <https://arxiv.org/abs/2210.05114>.
- [2] R. E. Megginson, *An Introduction to Banach Space Theory*, Graduate Texts in Mathematics 183, Springer, 1998.